

Subject Description Form

Subject Code	AAE5101
Subject Title	Next Generation Air Traffic Control and Air Traffic Flow Management
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with 1. Essential knowledge of airport, air traffic control and air traffic flow management; 2. The latest development of the Next Generation Air Transportation System (NextGen) and Asia-pacific airport collaborative decision-making (A-CDM); 3. Critical thinking training through case studies of NextGen from different countries.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. apply techniques to optimise the airport and air traffic capacity; b. understand and establish a review on the effectiveness of an air traffic management system; c. streamline airport, ground and air traffic operations to gain overall turn-a-round efficiency; and d. identify the airline-airport conflict resolution approach and risk management.
Subject Synopsis/ Indicative Syllabus	Overview of NextGen ATM System: An introduction to ATM systems and operations. Introduce definition, goal, and benefits of NextGen initiative on different aspects including airport, air traffic control and traffic flow management. Before and after NextGen: Provide a comparison of improved system before and after NextGen on communications, navigation, surveillance, air traffic control, etc. Air traffic control and management: Technical improvement in NextGen on Air traffic management, congestion control and capacity management, aviation system; Air traffic control and air traffic control aids; Seamless air traffic management and air navigation service; Extreme weather operations; Runway scheduling and capacity analysis: Runway capacity analysis; Airport airside and landside structure and layout; First-come first-served heuristics; Runway design and configuration. Operationalized NextGen: how to implement NextGen at the right places to raise the usage of deployed capabilities to achieve maximum

	benefits. Trajectory Based Operations (TBO), air and ground information exchange etc. Uncertainty quantification and risk assessment in ATM and ground operations: Recognizing frequent aircraft routes; Ground vehicles trajectory prediction and collision risk assessment. Aviation incident risk prediction; Risk assessment, anomaly detection, and uncertainty quantification in take-off and landing. NextGen initiatives from different countries and global harmonization: Introduce different focus of NextGen system by different countries, with detailed programs or strategies proposed and implemented. U.S.-EU collaboration in a global context for NextGen System. Advancement in airspace technology and performance indicators: Measurement of system performance; Key issue in airport collaborative decision making in Asia pacific; Critical elements of the Next Generation Air Transportation System (NextGen); Performance and concerns of the NextGen; Airspace Technology Demonstration (ATD): ATD-2/ATD-3. Future vision of airspace system: The future Airspace System need to accommodate emerging users and innovative business models for aviation and aerospace.																			
Teaching/Learning Methodology	Teaching is conducted through lectures and case study. Both the basic knowledge and theoretical models are going to be introduced. The understanding of how to address problems by using scientific tools is emphasised. Normally, examples of problem-solving techniques are taught in class and related scenarios are provided to students to enhance their application abilities. Aviation system robustness and resilience <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Case Study</td><td>√</td><td>√</td><td>√</td><td></td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√	√	√	Case Study	√	√	√	
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	a	b	c	d																
Lecture	√	√	√	√																
Case Study	√	√	√																	

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Assignment	30%	√	√		√
	2. Case study	40%	√	√	√	√
	3. Individual essay	30%		√	√	
	Total	100%				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 1.0 × Continuous Assessment The continuous assessment (100%) is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus via reading assignment and case study. Individual essay is used to assess the students’ capacities of self-study and problem-solving and understanding on a specific topic to fulfil the requirements of working in the aviation industry.					
Student Study Effort Expected	Class contact:					
	▪ Lecture/Case Study					39 Hrs.
	Other student study effort:					
	▪ Literature review/case study/reading					36 Hrs.
	▪ Self-learning/preparation					36 Hrs.
	Total student study effort					111 Hrs.
Reading List and References	1. Ashford, N. J., Stanton, H. M., Moore, C. A., Pierre Coutu, A. A. E., & Beasley, J. R. (2013). Airport operations. McGraw-Hill Education. 2. Cusick, S. K., Cortes, A. I., & Rodrigues, C. C. (2017). Commercial aviation safety. McGraw-Hill Education. 3. De Neufville, R., Odoni, A. R., Belobaba, P. P., & Reynolds, T. G. (2013). Airport systems: Planning, design, and management. McGraw-Hill Education. 4. Horonjeff, R., McKelvey, F. X., Sproule, W. J., & Young, S. B. (2010). Planning and design of airports. McGraw-Hill Education. 5. Wells, A. T. (2007). Air transportation: A management perspective: Ashgate Publishing, Ltd. 6. Young, S. B., & Wells, A. T. (2011). Airport planning and management. McGraw-Hill Education.					